

ClubGear

A Campus Club Equipment Booking
and Management System

Short Project Report

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Programme **B.Tech CSE, Second Year**

Course **Java Programming**

Repository <https://github.com/suyash25bai11183-ui/Java-Project>

1. Introduction

College clubs share equipment — cameras, microphones, projectors, tripods and lighting gear — that is usually tracked in a paper register or a chat group. This causes double bookings, lost items, unnoticed late returns, and damaged equipment being handed out again. **ClubGear** is a console-based Java application that replaces that register: every club, member and equipment item gets a unique ID, and the system automatically enforces the booking rules that a manual process cannot.

1.1 Objectives

- Track clubs, members and equipment with unique auto-generated IDs
- Process bookings through a defined life cycle and reject conflicting requests
- Detect overdue bookings automatically against the current date
- Record equipment condition on return and manage a maintenance history
- Persist all data in plain text files with no database
- Demonstrate core Java: OOP, collections, enums, exceptions, dates and file I/O

1.2 Technology stack

Item	Detail
Language	Java 17, compiled with <code>--release 17</code>
Collections	<code>ArrayList</code> , <code>HashMap</code> , <code>HashSet</code>
Dates	<code>java.time.LocalDate</code> , <code>java.time.LocalDateTime</code>
Storage	Five plain CSV text files, no database
Testing	Self-written 55-test suite + 22-case manual test plan
External libraries	None

2. Functional Modules

The system is built from seven functional modules, each mapped to specific menu options.

#	Module	Capabilities
1	Club management	Add, view, search clubs; reject duplicate names
2	Member management	Register members; validate and de-duplicate emails
3	Equipment management	Add equipment; track quantity, condition, status
4	Equipment booking	Create bookings; detect date and status conflicts
5	Booking status management	Approve, cancel, issue, return; detect overdue
6	Return and maintenance	Record condition, log damage, repair cycle
7	Reports	Totals, availability, overdue list, usage statistics

2.1 Business rules enforced

1. Club names must not be duplicated
2. Member emails must not be duplicated
3. Equipment quantity must be greater than zero
4. Damaged or under-maintenance equipment cannot be booked
5. A member cannot book the same item for overlapping dates
6. Two bookings for the same item cannot overlap
7. Return date cannot be earlier than the start date
8. Only approved bookings can become active
9. Only active (or overdue) bookings can be returned
10. Only returned or cancelled bookings release the equipment
11. Overdue bookings are detected using the current date
12. Invalid menu input must not crash the program

3. System Architecture

ClubGear follows a three-layer design: presentation (menu and input), domain logic (business rules), and persistence (CSV read/write). Each layer talks only to the layer directly below it.

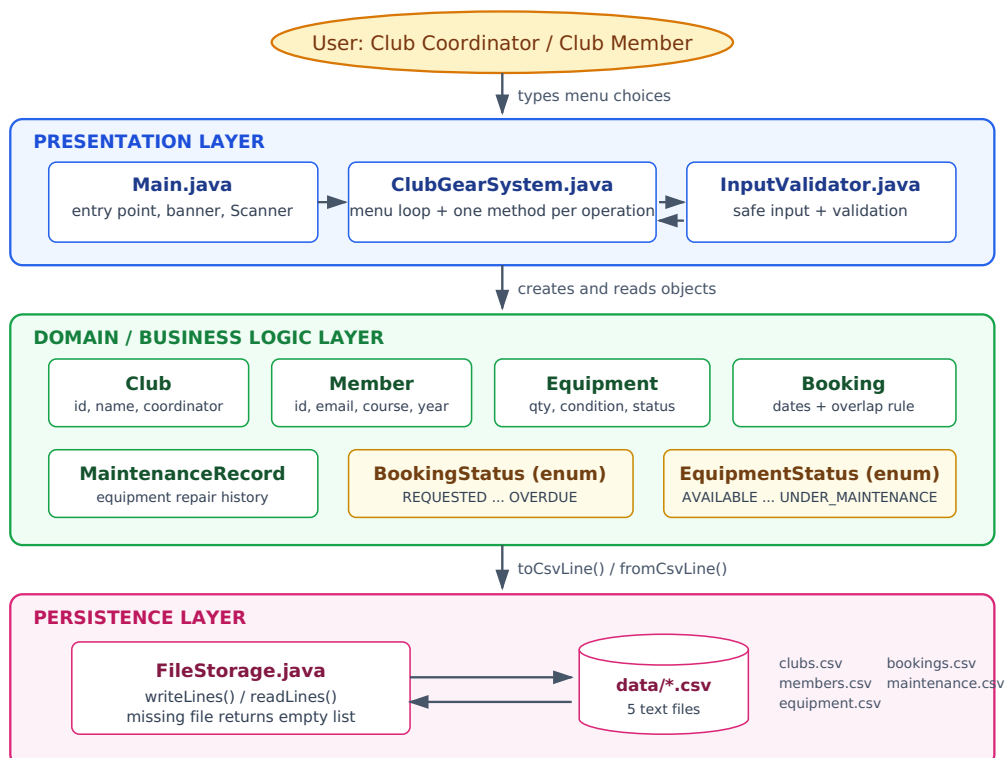


Figure 1 — Three-layer architecture of ClubGear

4. Implementation Summary

Class	Responsibility
Main	Entry point; creates the Scanner and starts the system
ClubGearSystem	Menu loop and one method per operation
Club , Member , Equipment	Core data records with CSV conversion
Booking	Booking record, including the date-overlap rule
MaintenanceRecord	One line of repair history per equipment item
BookingStatus , EquipmentStatus	Enums controlling valid state transitions
FileStorage	CSV read/write, missing-file handling
InputValidator	Safe, re-prompting keyboard input

4.1 Key snippet — clash detection

```
private Booking findClashingBooking(String equipmentId, LocalDate start, LocalDate end) {
    for (Booking booking : bookings) {
        if (booking.getEquipmentId().equals(equipmentId)
            && booking.isBlockingEquipment()
            && booking.overlapsWith(start, end)) {
            return booking;
        }
    }
    return null;
}
```

5. Testing

Type	Coverage
Automated (ValidationTests.java)	55 tests: date overlap, status rules, quantity, email validation, CSV round-trips, damaged file lines, file storage
Manual (TestPlan.md)	22 test cases: duplicates, conflicts, approvals, returns, overdue detection, save/load

```
Total tests : 55
Passed      : 55
Failed      : 0
RESULT: ALL TESTS PASSED
```

Result. The project compiles cleanly with `javac --release 17 -Xlint:all` (zero errors, zero warnings) and all 55 automated tests pass.

6. Sample Output

```
===== BOOKING CONFIRMATION =====
Booking ID   : BKG004
Member      : Rahul Nair (MEM003)
Equipment   : Camera Tripod (EQP004)
Club        : Photography Club
From        : 2026-09-20
Until       : 2026-09-22
Status      : REQUESTED
=====

Error: this equipment is already booked from 2026-09-20 to 2026-09-22 (booking BKG004).
Error: this equipment is DAMAGED and cannot be booked right now.
```

7. Conclusion and Future Enhancements

ClubGear meets its objectives: it tracks clubs, members and equipment; it processes bookings through a full life cycle while enforcing twelve business rules; it detects overdue returns automatically; and it persists all data without a database. The project demonstrates practical use of core Java — classes, encapsulation, enums, collections, exception handling, date/time APIs and file I/O — within a scope appropriate for a second-year course project.

Future enhancements

- Role-based login for coordinators and members
- Real stock counting instead of one-record-per-asset
- Email/SMS return reminders and late-return fines
- Replacing CSV storage with SQLite via JDBC
- A JavaFX or Swing graphical front end over the existing classes

For full requirements, UML diagrams, design rationale, complete screenshots and the detailed test log, see [ClubGear_Project_Report.pdf](#) and the `docs/` folder in the repository.

